JAWAHARLAL NEHRU NEW COLLEGE OF ENGINEERING
Department of Computer Science & Engineering
Shivamogga – 577204

## Internship Organization : Rooman Technologies

## Data Analytics Intern

# Business Insights Dashboard using Superstore Dataset

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# 1.Problem Statement

Businesses generate large volumes of sales data on a daily basis, but extracting meaningful insights from this data is often challenging. Traditional reporting methods mainly focus on past data and fail to clearly highlight trends, loss areas, or key contributors to revenue. As a result, decision-makers find it difficult to understand business performance and take effective actions.

# 2.Objectives

- To analyze sales and profit data from the Superstore dataset
- To identify trends and patterns in business performance
- To understand the impact of discount on profitability
- To compare performance across regions and categories
- To identify top-performing products and key revenue contributors
- To create interactive dashboards for better data visualization
- To support data-driven decision-making using insights

# 3.Dataset Overview

- Superstore dataset containing retail sales data
- Includes features like Sales, Profit, Discount, Quantity
- Covers Regions, Categories, and Sub-Categories
- Contains Order Date for time-based analysis
- Used to analyze overall business performance

# 4.Tech Stack

- Python (Jupyter Notebook) for data analysis
- Pandas and NumPy for data processing
- Matplotlib and Seaborn for visualization
- Power BI for interactive dashboard creation
- React.js for advanced analytical dashboard

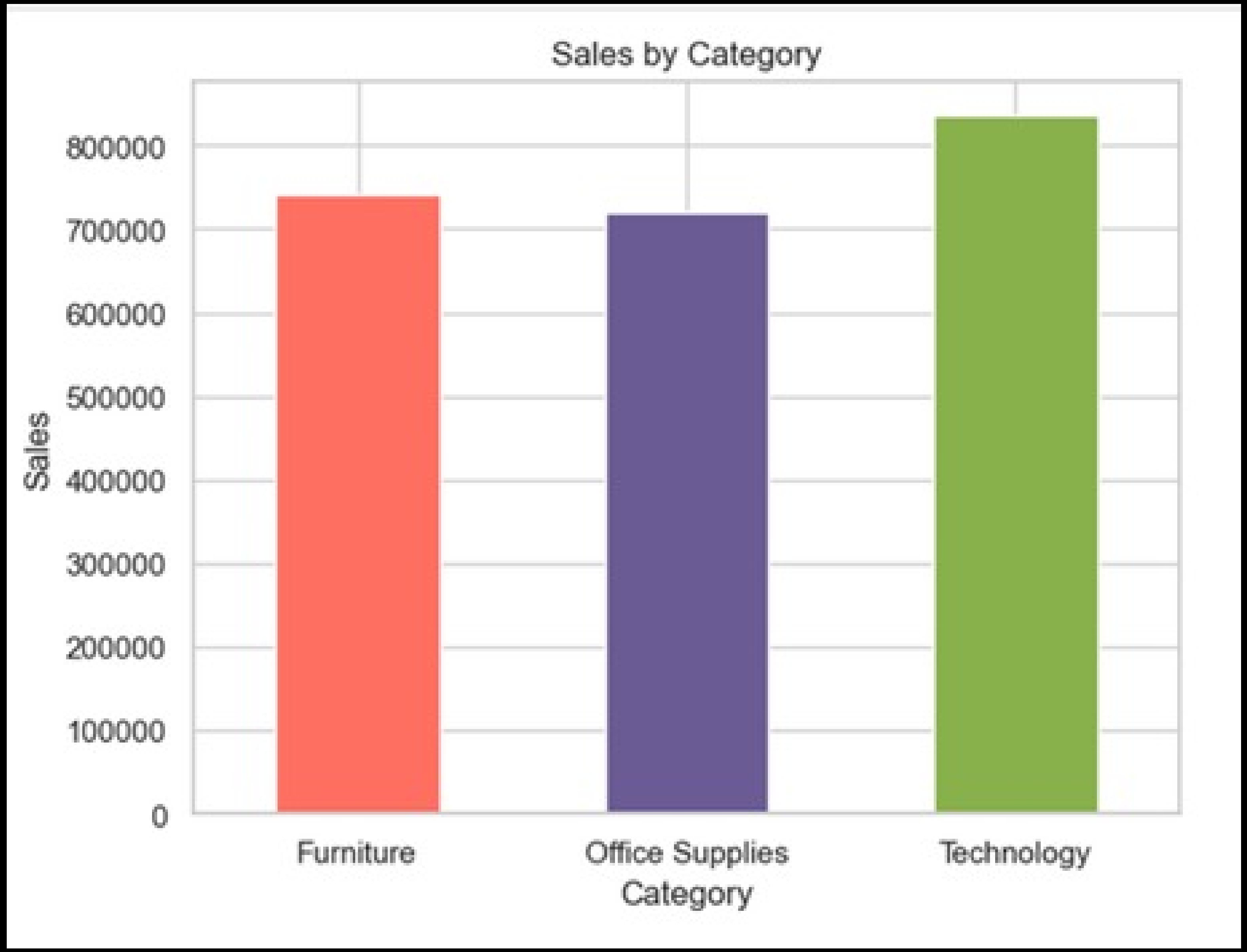
# 5.Data Cleaning

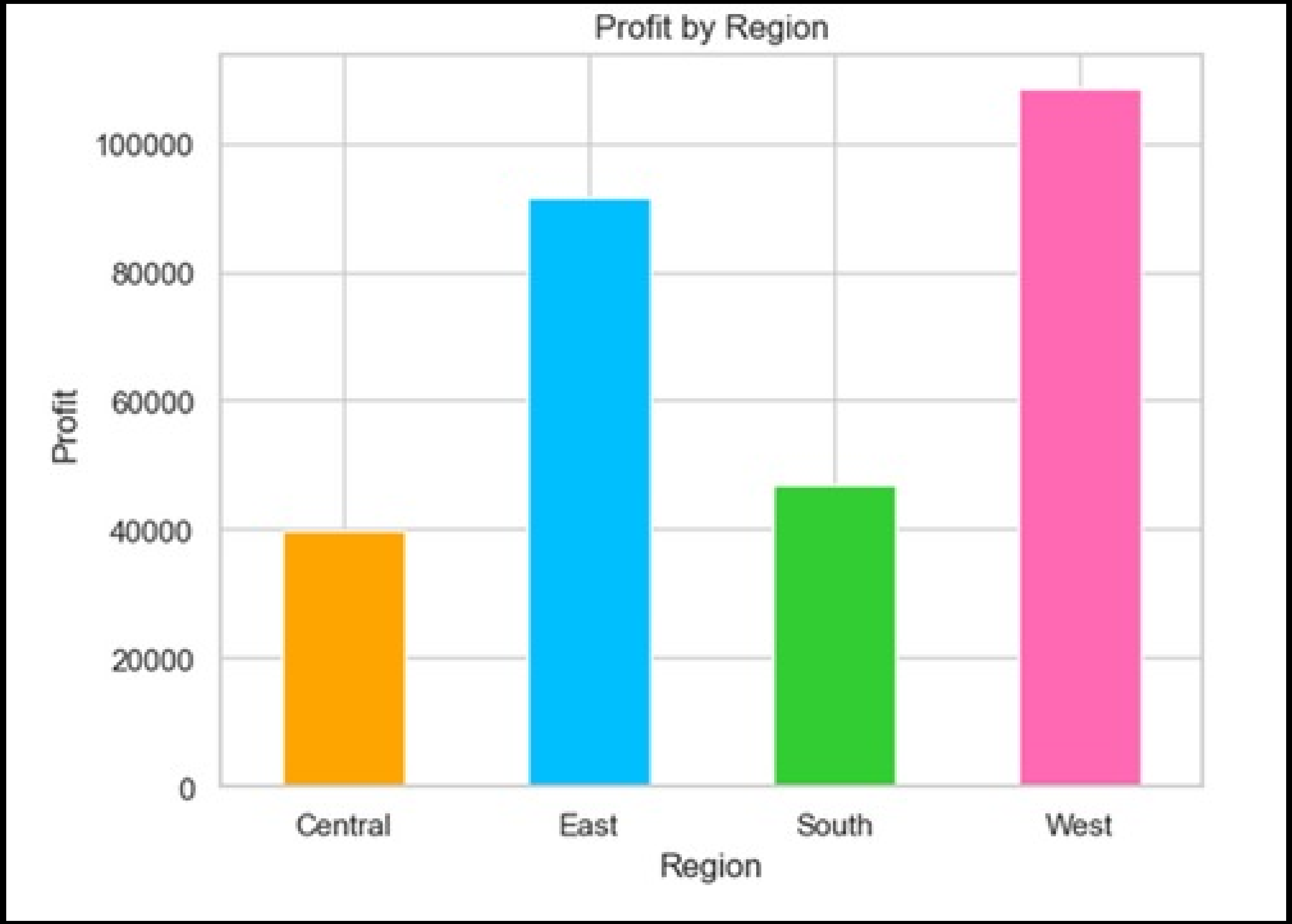
- Converted Order Date column into datetime format
- Verified dataset structure (shape, columns, data types)
- Checked for missing (null) values
- Removed duplicate records
- Prepared dataset for analysis

# 6.Analysis Process

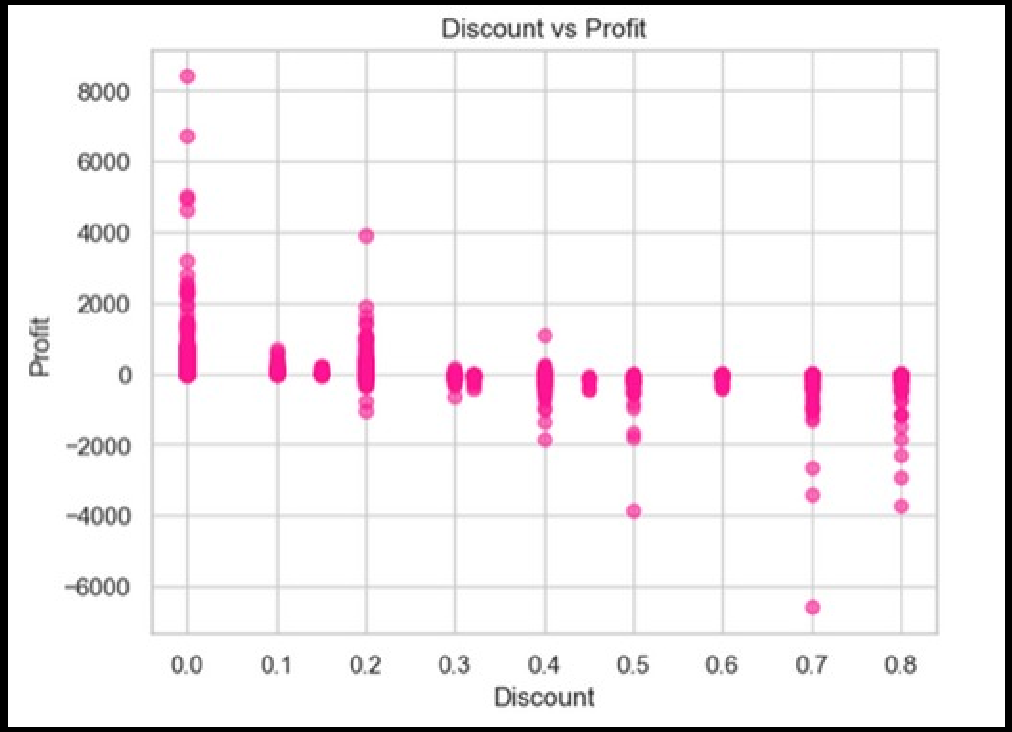
- Performed Exploratory Data Analysis (EDA)
- Analyzed sales and profit distribution
- Compared performance across regions and categories
- Studied time-based sales trends
- Identified top products and key contributors
- Analyzed relationship between discount and profit

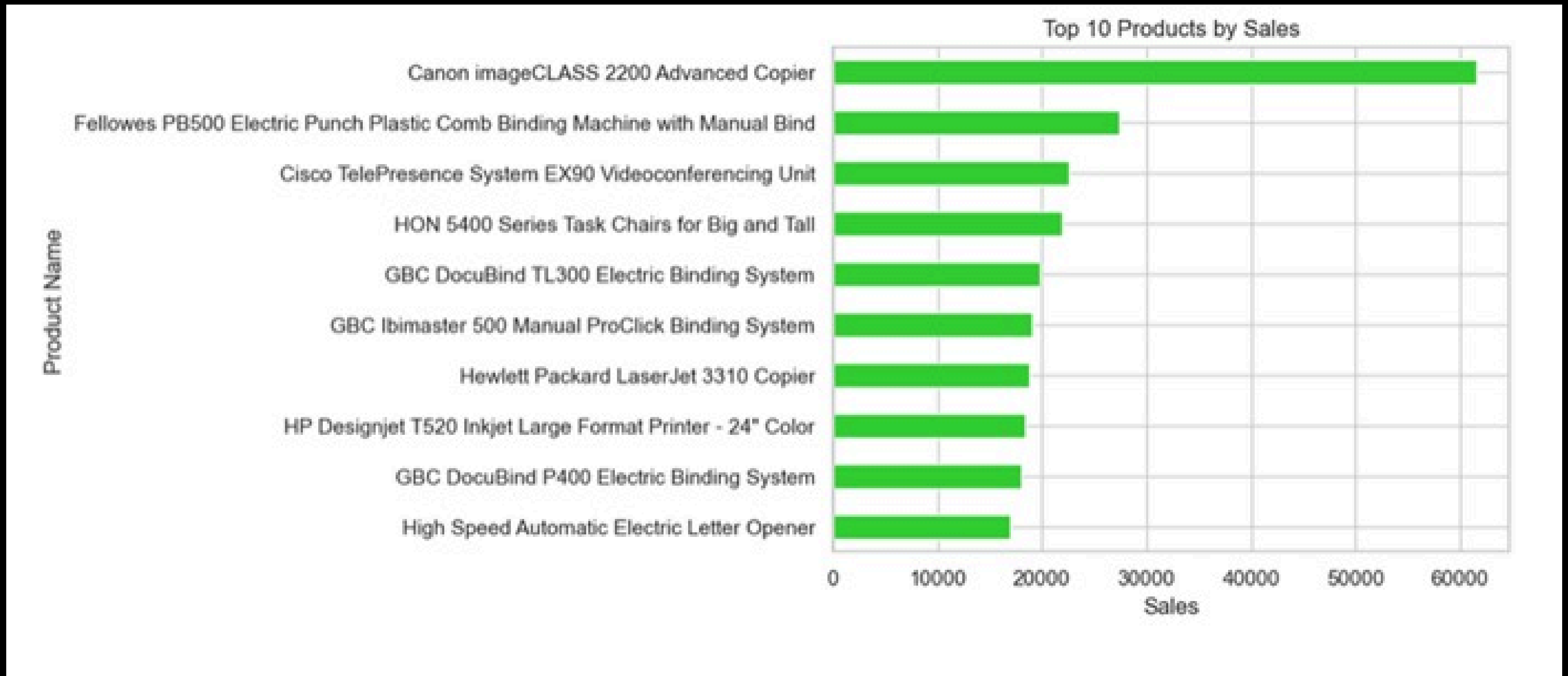
# 7.Business Insights through Visualizations

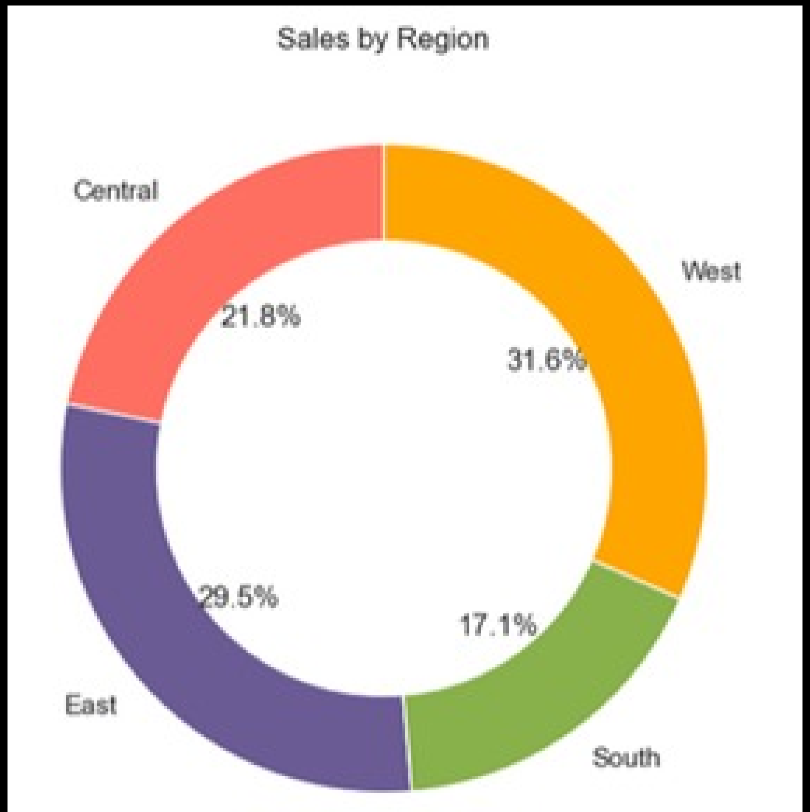


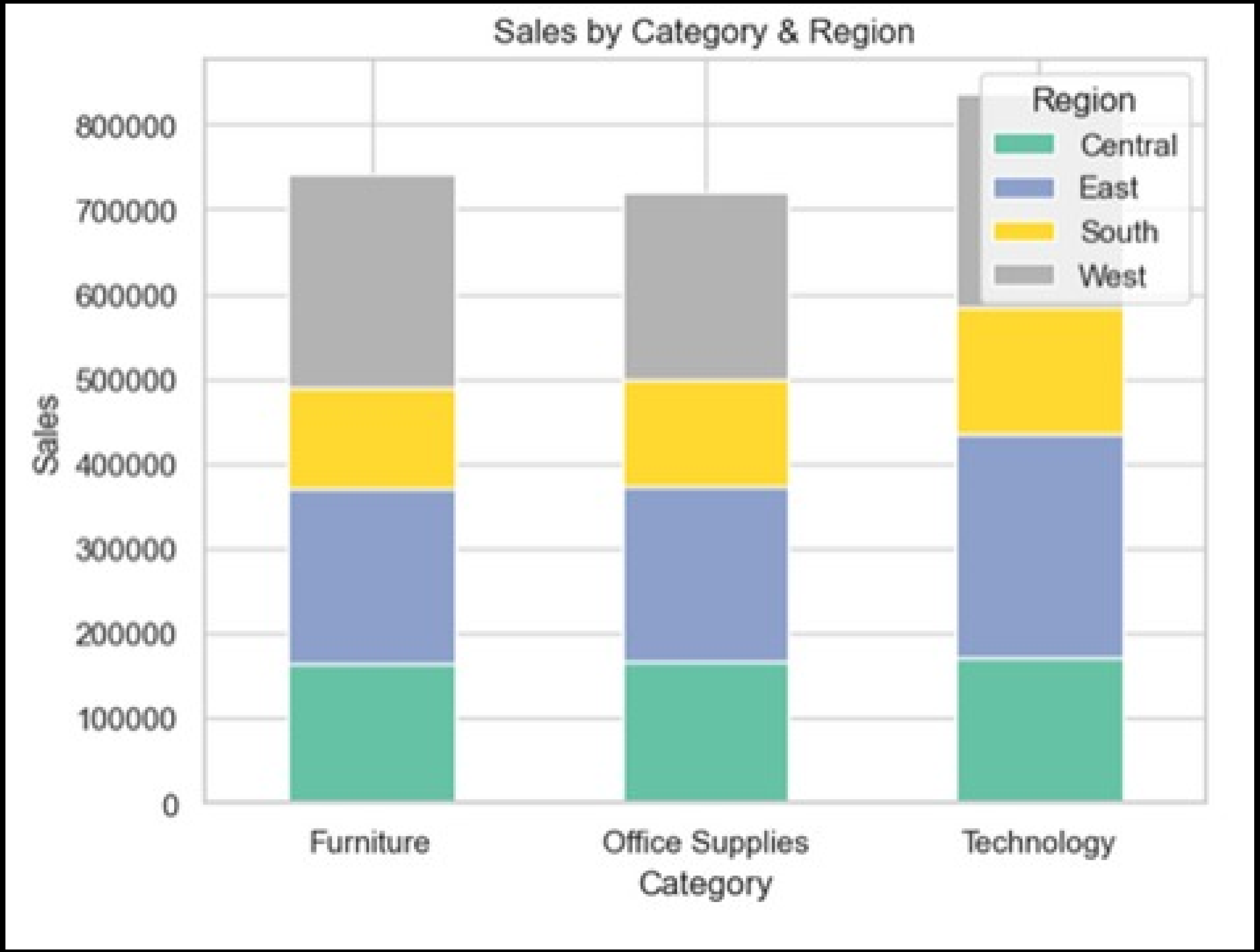
# 8.Data Analysis & Visual Insights

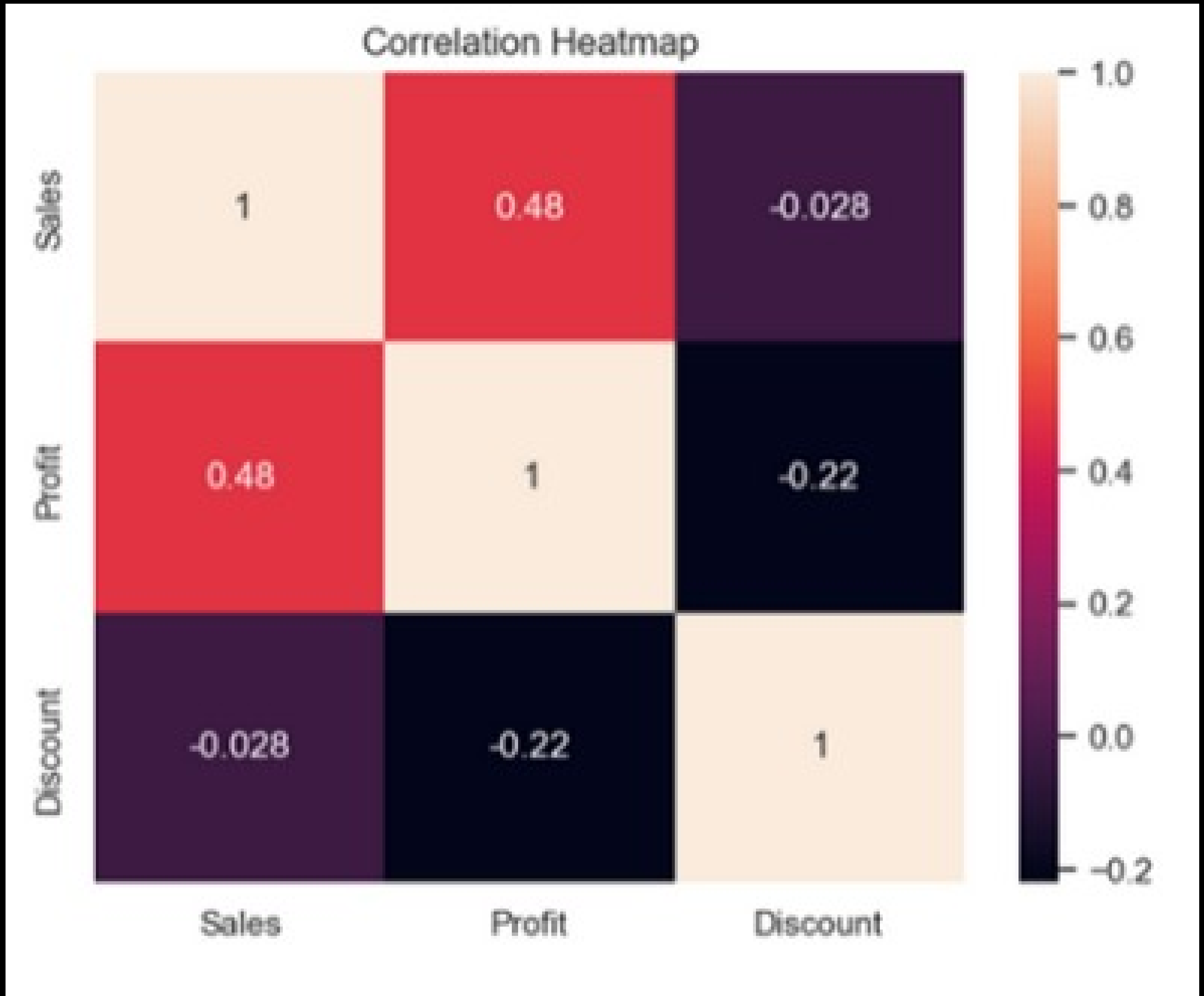



# 9.Visual Analysis using Jupyter Notebook



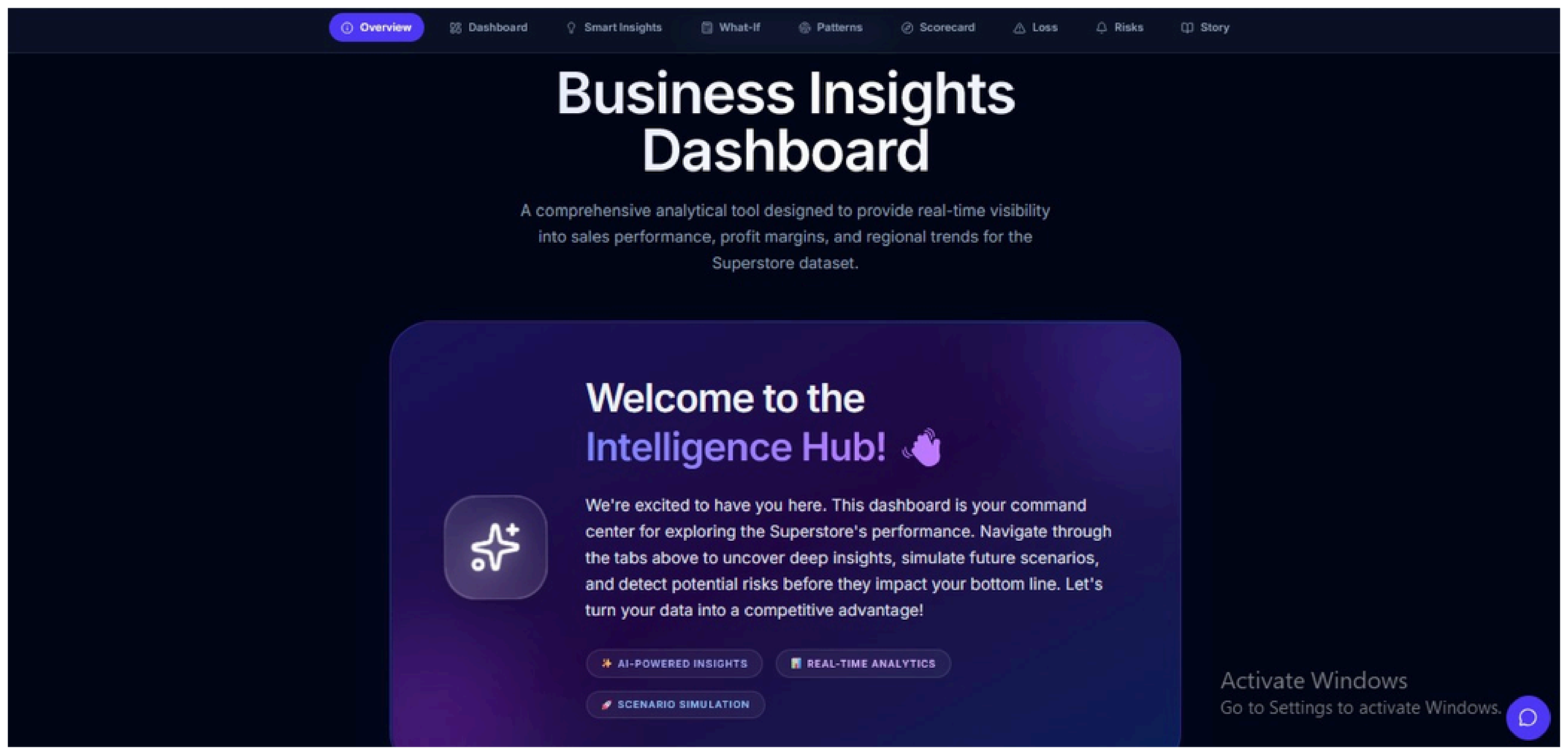
# 10.Visualization Insights

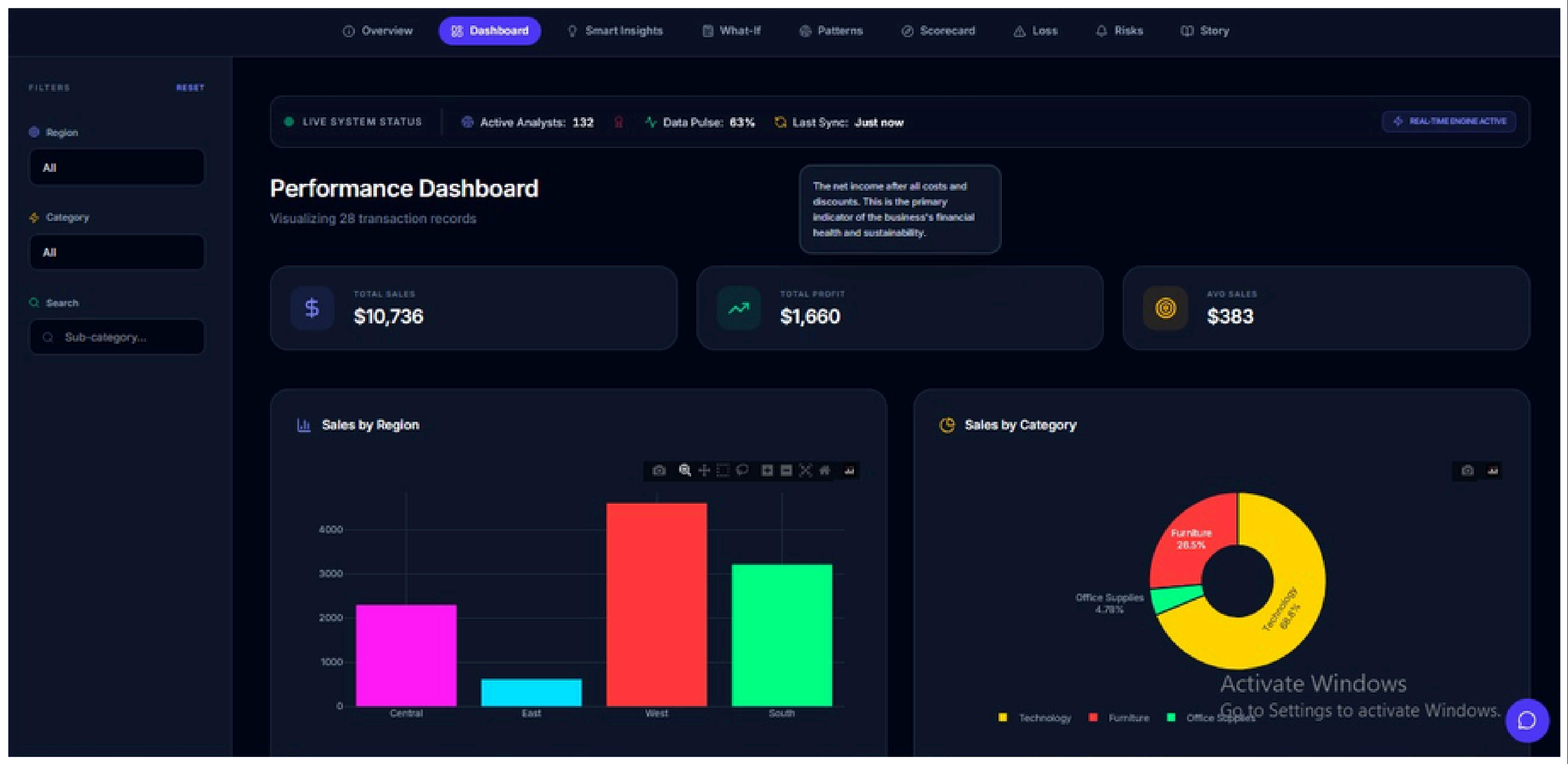
- **Sales by Category:** Technology category generates the highest sales among all categories
- **Profit by Region:** West region has the highest profit, while Central has the lowest
- **Monthly Sales Trend:** Sales fluctuate over time with higher values in later months
- **Discount vs Profit:** Higher discounts lead to lower profit and possible losses
- **Top Products by Sales:** A few products, especially Canon Copier, contribute major revenue
- **Sales by Region (Donut Chart):** West region contributes the highest share of total sales
- **Sales by Category & Region:** Technology performs strongly across all regions, especially West and East
- **Correlation Heatmap:** Sales and profit are positively related, while discount negatively affects profit

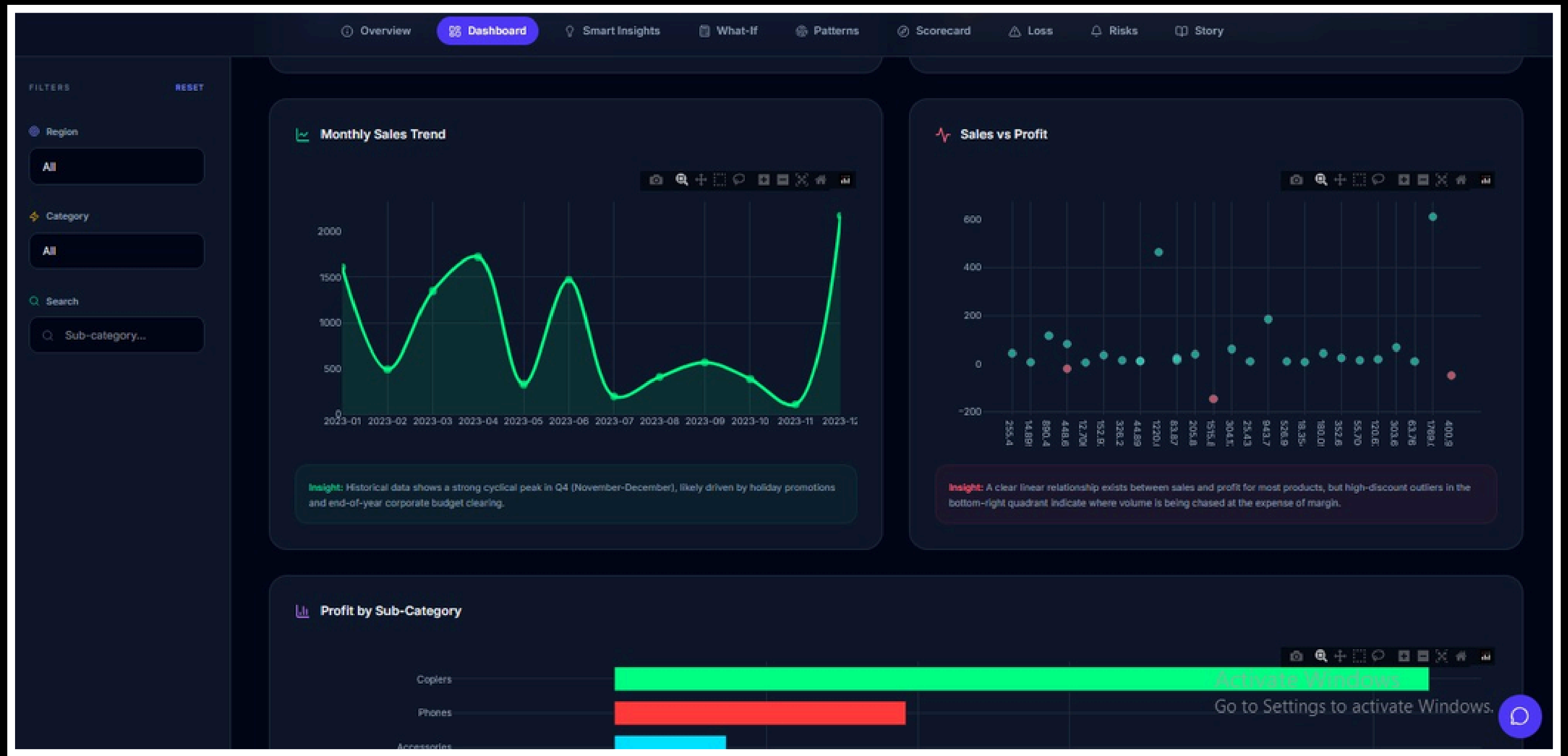
# 11.Business Insights Dashboard (Power BI)

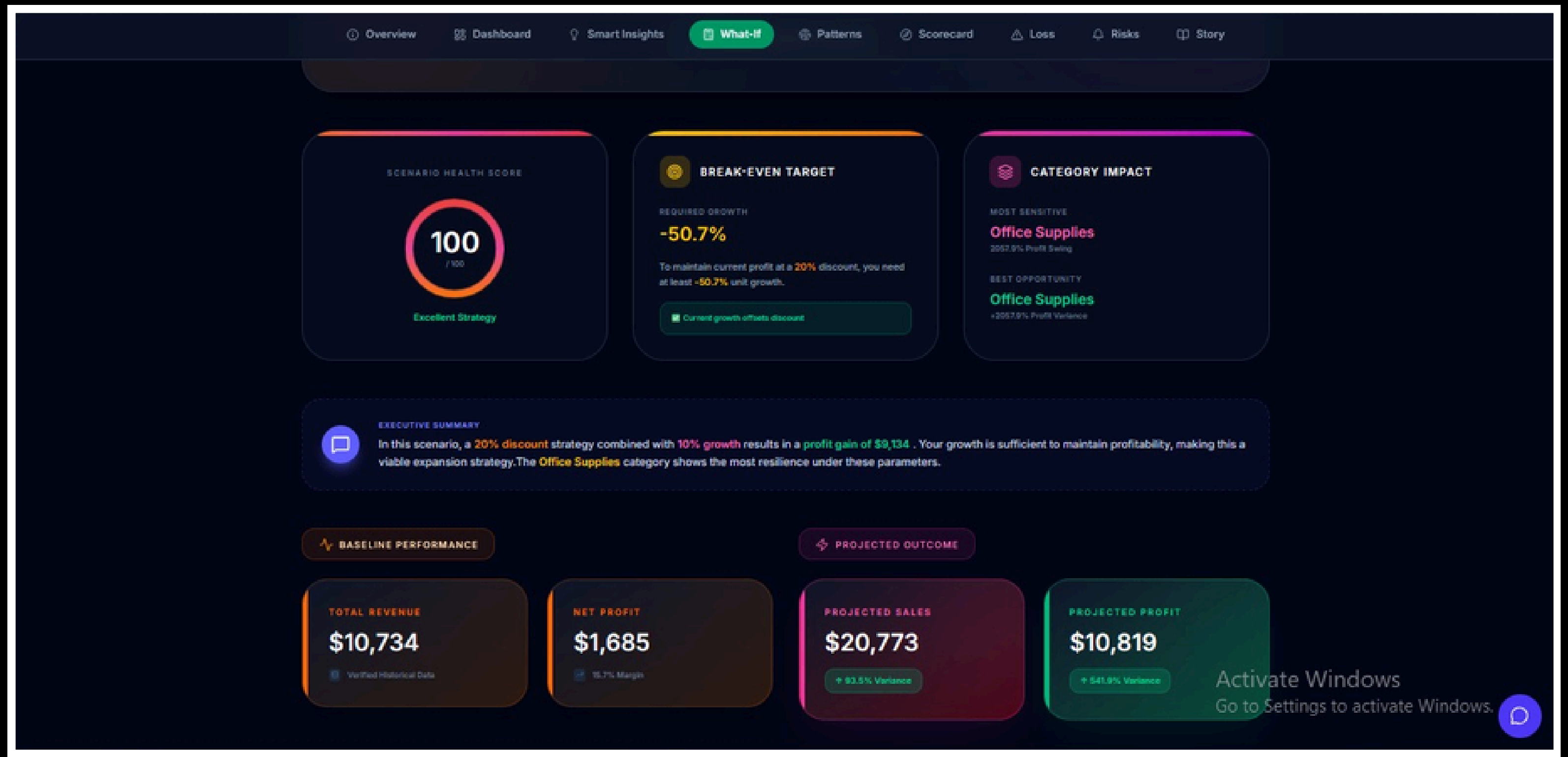
- The Power BI dashboard provides an interactive overview of business performance using key metrics such as total sales, profit, and quantity. It includes visualizations for category-wise sales, regional performance, customer contribution, and the impact of discounts on profitability.
- The dashboard also includes filters that allow users to explore data dynamically across different regions and categories. This helps in identifying trends, comparing performance, and making better business decisions.

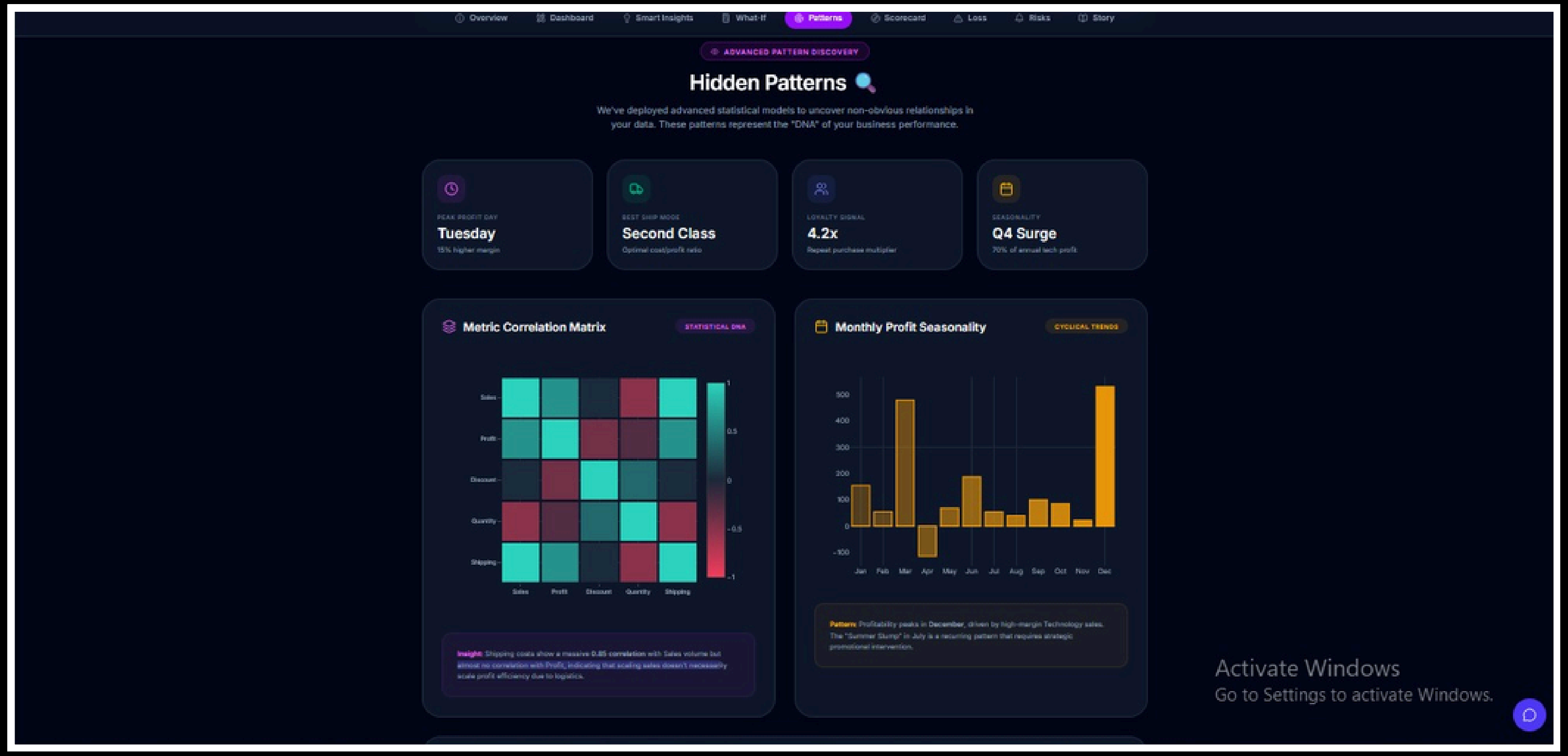
# 11.Business Insights Dashboard (React)

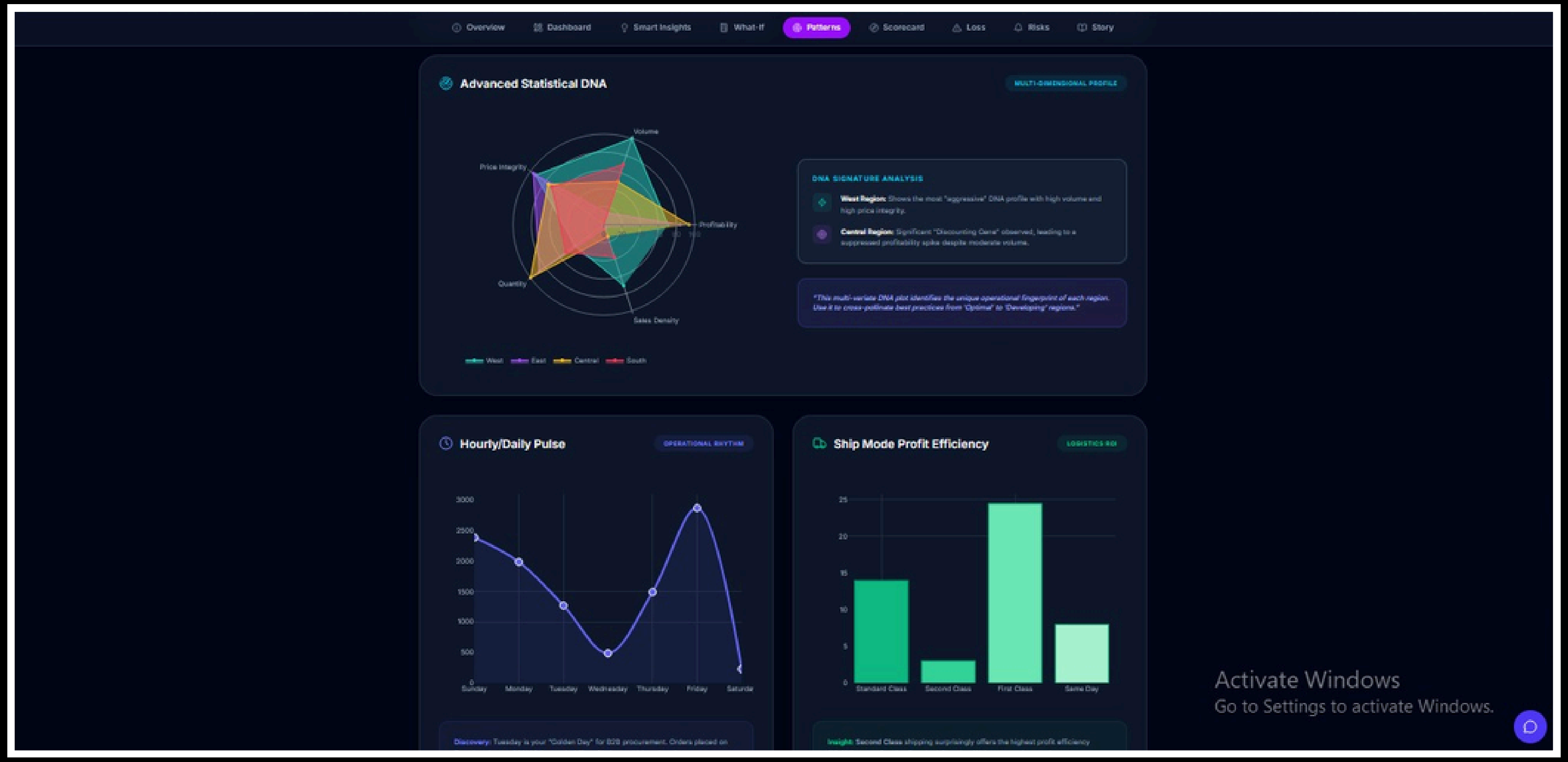


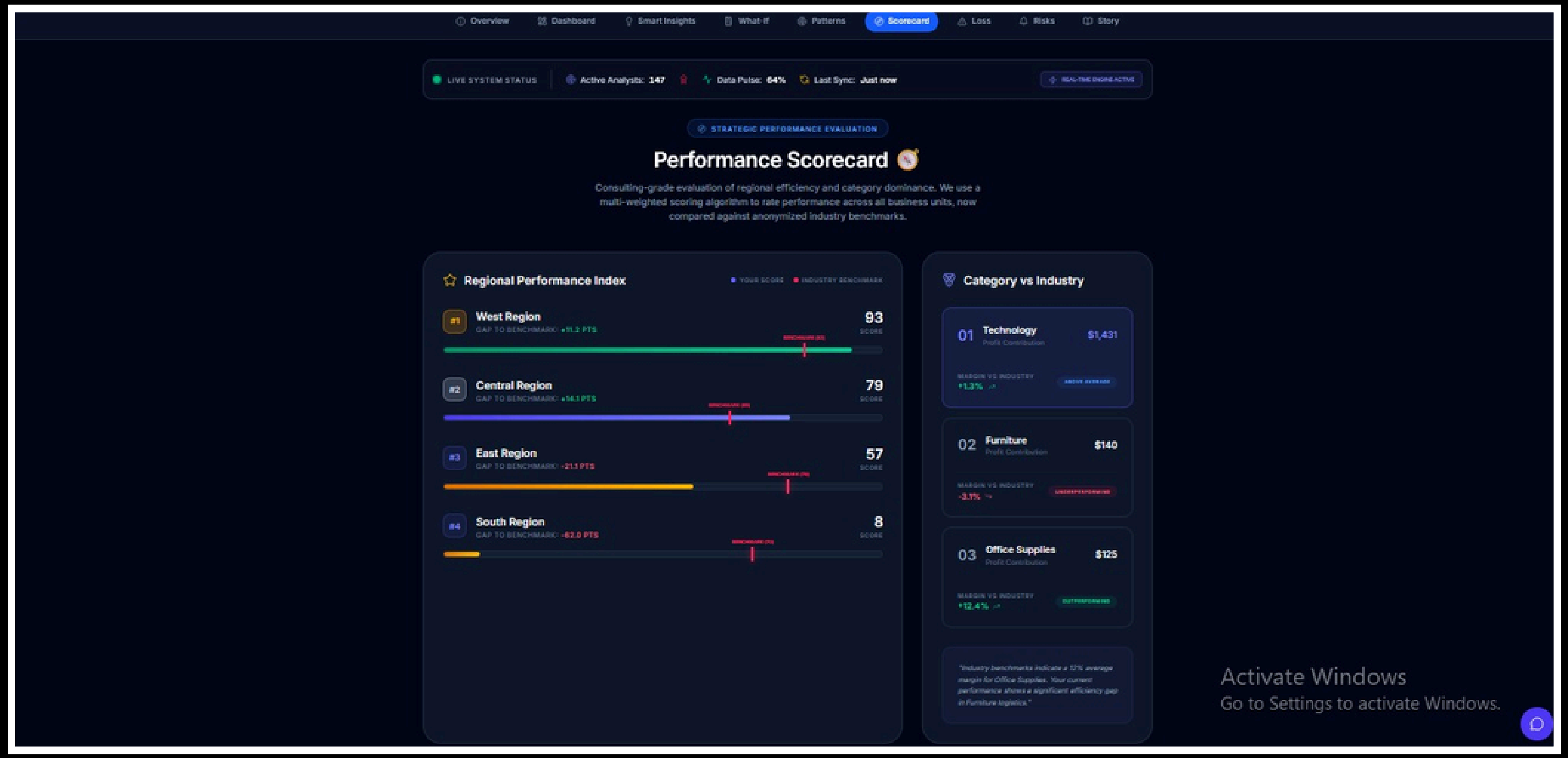



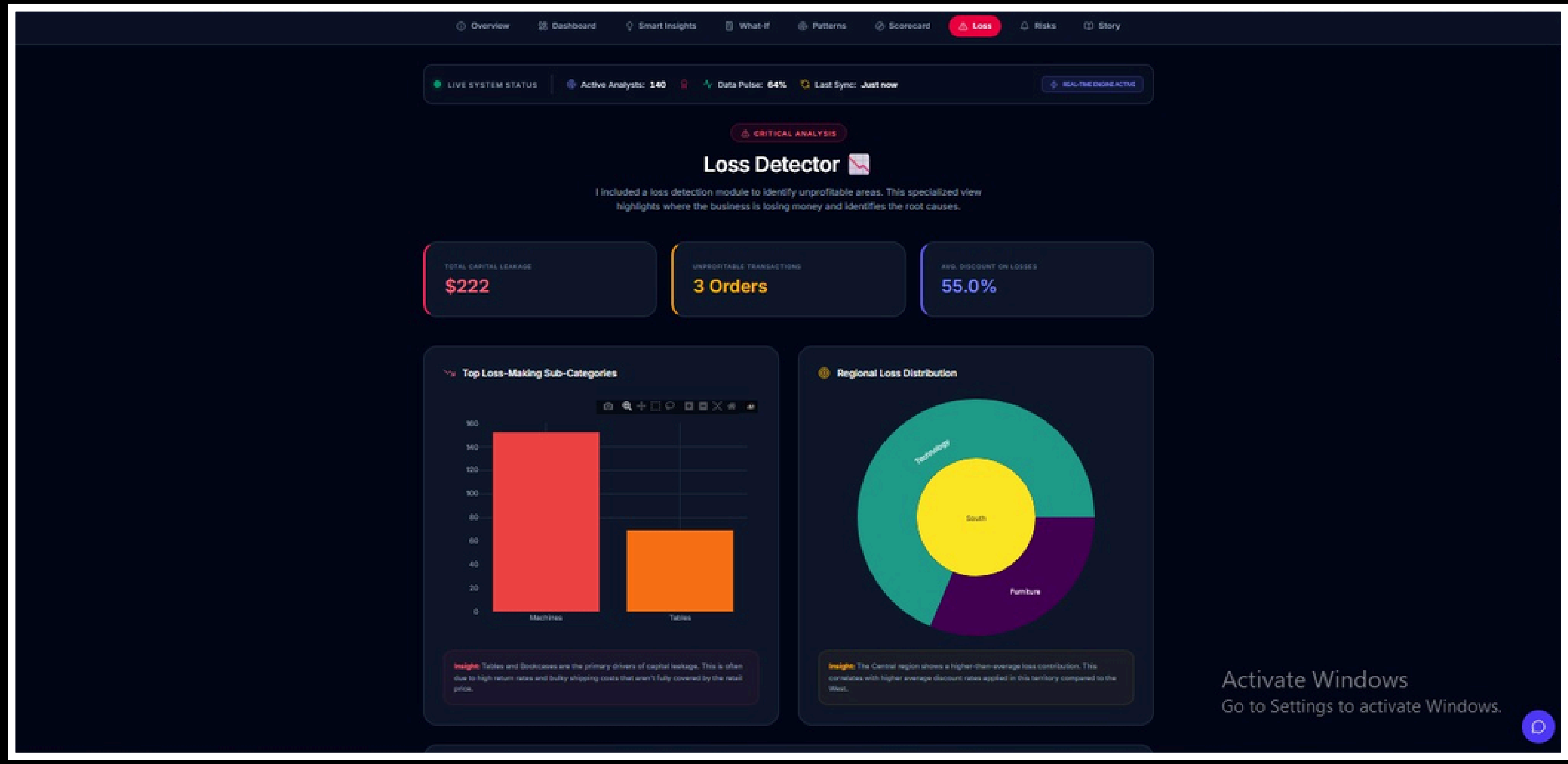




Overview
Dashboard
Smart Insights
What-If
Patterns
Scorecard
Loss
Risks
Story
ADVANCED PATTERN DISCOVERY
Hidden Patterns
We've deployed advanced statistical models to uncover non-obvious relationships in your data. These patterns represent the "DNA" of your business performance.
Tuesday
Second Class
4.2x
Q4 Surge
Metric Correlation Matrix
Monthly Profit Seasonality
Activate Windows
Go to Settings to activate Windows.

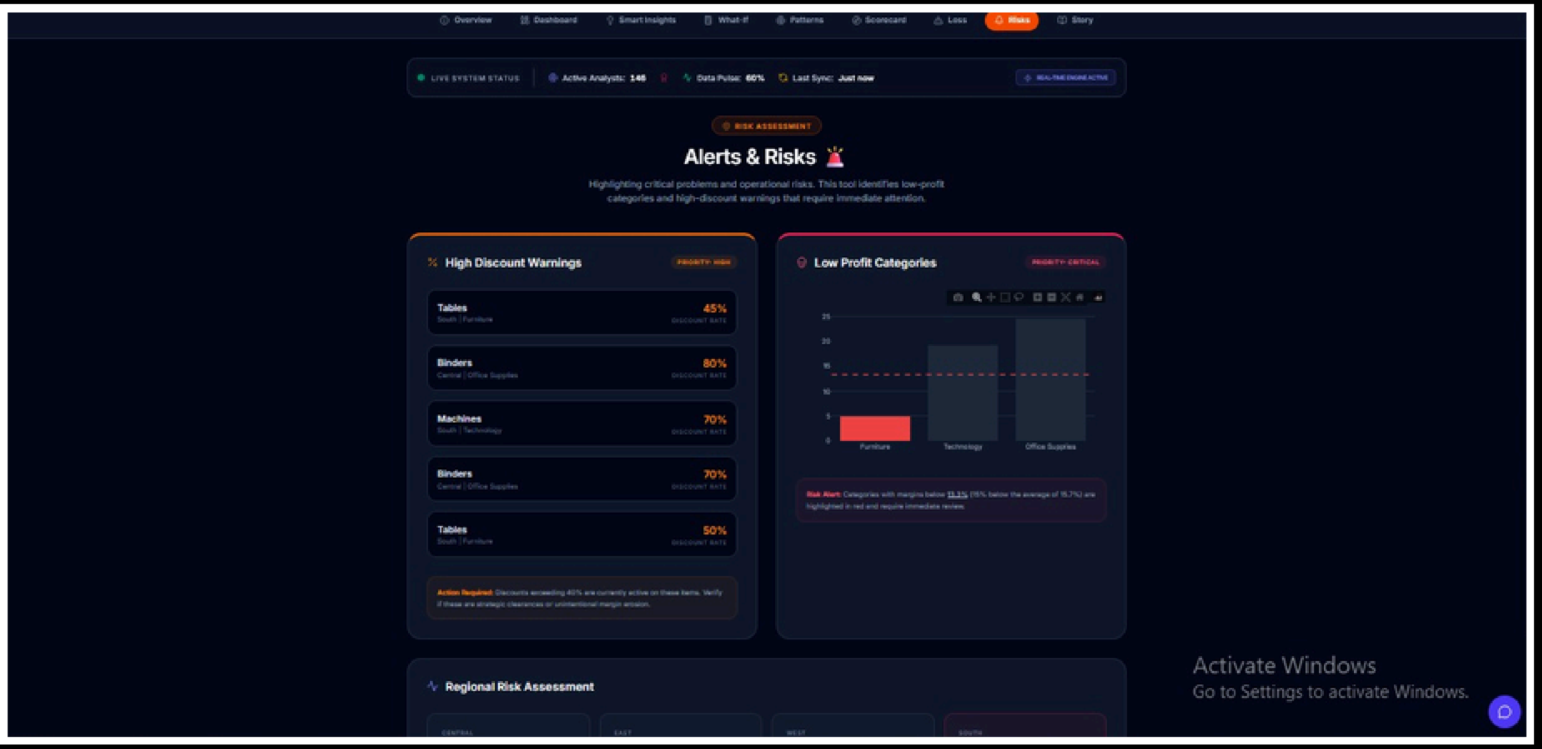

Overview
Dashboard
Smart Insights
What-If
Patterns
Scorecard
Loss
Risks
Story
Advanced Statistical DNA
Hourly/Daily Pulse
Ship Mode Profit Efficiency
Activate Windows
Go to Settings to activate Windows.

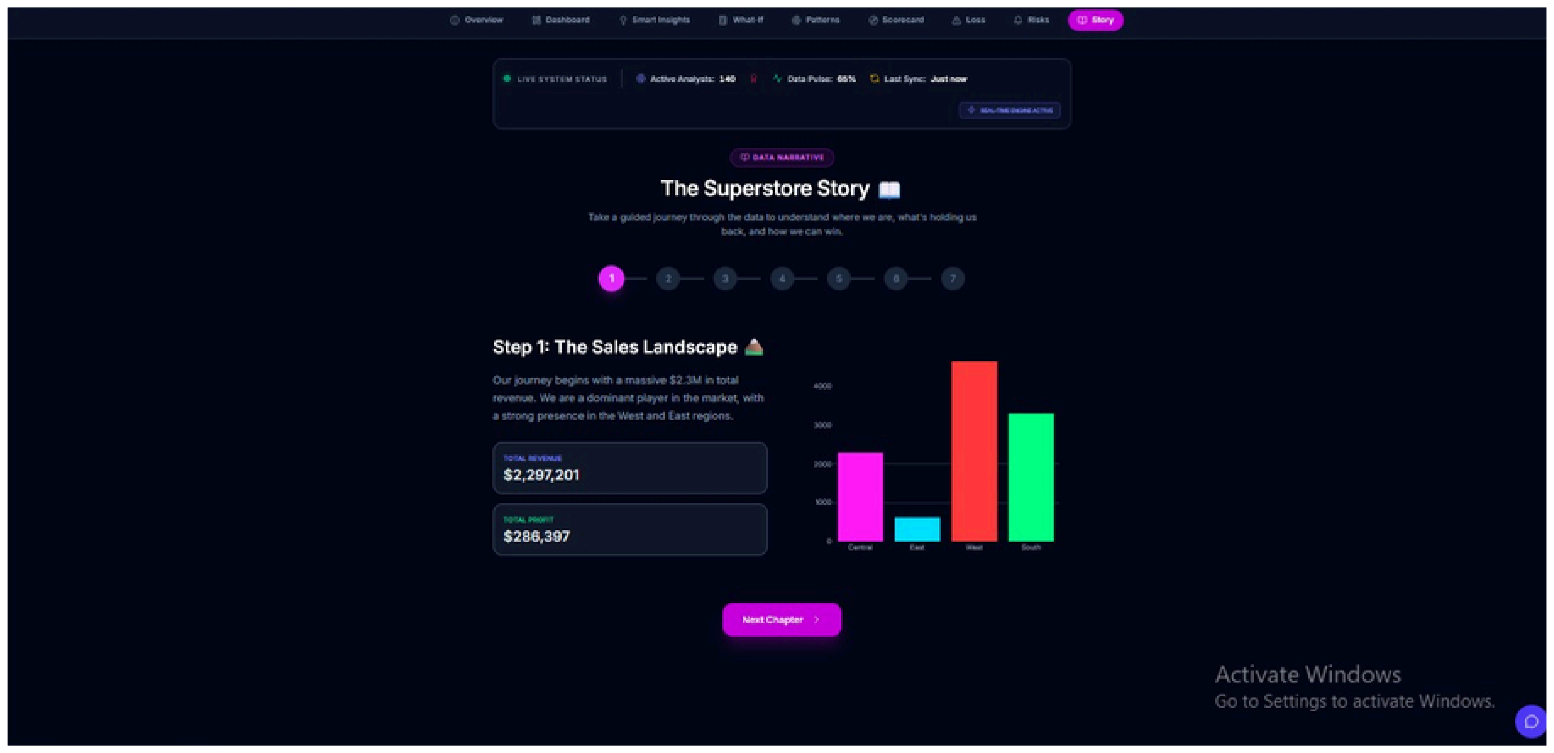

Overview
Dashboard
Smart Insights
What-If
Patterns
Scorecard
Loss
Risks
Story
Performance Scorecard
Consulting-grade evaluation of regional efficiency and category dominance. We use a multi-weighted scoring algorithm to rate performance across all business units, now compared against anonymized industry benchmarks.
Regional Performance Index
West Region
93
Central Region
79
East Region
57
South Region
8
Category vs Industry
01 Technology $1,431
02 Furniture $140
03 Office Supplies $125
Activate Windows
Go to Settings to activate Windows.


Overview
Dashboard
Smart Insights
What-If
Patterns
Scorecard
Loss
Risks
Story
Loss Detector
I included a loss detection module to identify unprofitable areas. This specialized view highlights where the business is losing money and identifies the root causes.
$222
3 Orders
55.0%
Top Loss-Making Sub-Categories
Regional Loss Distribution
Activate Windows
Go to Settings to activate Windows.

# React Dashboard Tabs Overview

- **Overview:** Provides a quick summary and navigation to all dashboard sections
- **Dashboard:** Displays key metrics like sales, profit, and performance insights
- **Smart Insights:** Highlights important patterns and relationships in the data
- **What-If Analysis:** Allows users to test different scenarios and predict outcomes
- **Hidden Patterns:** Shows deeper relationships and trends within the data
- **Performance Scorecard:** Evaluates and compares performance across regions
- **Loss Detector:** Identifies areas where profit is being lost
- **Alerts & Risks:** Highlights critical issues and high-risk areas
- **Story Mode:** Presents insights in a simple and easy-to-understand narrative



# 12.Results

- The analysis of the Superstore dataset revealed several important business insights. The Technology category was identified as the highest revenue-generating category, indicating strong demand for technology products. Regional analysis showed that the West region contributes the highest sales and profit, making it the top-performing region.

- The study also highlighted that higher discounts negatively impact profitability, with many high-discount transactions resulting in losses. Additionally, it was observed that a small number of products contribute a major portion of total revenue, emphasizing the importance of key products. Time-based analysis showed variations in sales, indicating trends and possible seasonal patterns.

# 13.Learning Outcomes

- Gained understanding of data cleaning and preprocessing techniques
- Learned Exploratory Data Analysis (EDA) to identify patterns and trends
- Improved data visualization skills using charts and graphs
- Developed interactive dashboards using Power BI
- Built an advanced dashboard using React for better analysis
- Understood how data analytics supports real-world business decisions

# 14.Project Links

**1.GitHub (Notebook + Power BI):**

https://github.com/akshata130504/Business-Insights-Dashboard-using-Superstore-Dataset

**2.React Dashboard Repository:**

https://github.com/akshata130504/Business-Insights-Dashboard-using-React-

**3.Live React Dashboard:**

https://business-insights-dashboa-git-43abc8-akshata-t-rathods-projects.vercel.app

# 15.Next Steps

- Integrate real-time data for continuous analysis
- Apply predictive analytics for future forecasting
- Enhance dashboard with more interactive features
- Improve business recommendations using advanced models

# 16.Conclusion

- This project successfully demonstrates how raw business data can be transformed into meaningful insights using data analysis and visualization techniques. By analyzing sales, profit, and discount patterns, key trends and performance areas were identified.

- The dashboards provide an interactive and clear view of business performance, helping in understanding trends, identifying problem areas, and supporting better decision-making. Overall, the project highlights the importance of data analytics in improving business strategies and outcomes.

- “This project shows how data analytics can help businesses make smarter and more effective decisions.”

# Thank You